

ATRIA

Dynamic Emergency Department Triage & Reassessment Workstation

Document field	Value
Version	1.0 — developer handoff
Date	27 August 2026
Status	Prototype specification; clinical validation required
Primary audience	Frontend, backend, ML, data, clinical informatics, security and QA
Jurisdiction assumption	United States; HIPAA-oriented prototype
Reference build	ATRIA_Intuitive_Flow_Deterioration_UI.html

Product promise

ATRIA continuously reorders attention in the emergency-department waiting room so that patients who are most urgent—or becoming more urgent—are seen sooner. It supports the triage nurse; it does not prescribe treatment and does not replace the nurse's final ESI sign-off.

Safety status

This PRD specifies a research/prototype implementation. The thresholds, reassessment intervals, model behavior and operational assumptions must be reviewed, approved and version-controlled by the deploying hospital's clinical governance body before any use with real patients.

Prepared as the engineering source of truth for turning the current offline dashboard into a functional, testable full-stack prototype.

Document map

This PRD is intentionally detailed. Product behavior, clinical safety rules, model boundaries, data contracts and acceptance tests are all specified so that engineering can implement without reverse-engineering the screen.

- 1–4: product intent, users, scope and success measures
- 5–8: experience architecture and functional behavior
- 9–12: clinical safety layers, ESI, queue ranking and reassessment
- 13–17: model contract, data model, APIs, integrations and operations

- 18–22: security, non-functional requirements, validation, rollout and risks
- Appendices: ESI nomenclature, prototype configuration, audit event schema and glossary

Decision summary

Decision	Product specification
Primary clinical task	Nurse independently selects ESI 1–5; ATRIA reveals its recommendation only afterward.
System authority	ATRIA may trigger deterministic red flags, abstain, recommend ESI and re-rank attention. The nurse owns final sign-off and disposition.
Queue safety	ESI acuity bands are strict. Operational load and waiting time may reorder patients within a band, not silently change clinical ESI.
Critical undertriage	A model-independent red-flag gate runs before ML. Critical physiology forces ESI 1 and top-of-queue visibility.
Ambiguity	Missing heart rate, blood pressure or oxygen saturation causes abstention and a mandatory human check; missingness is never treated as normal.
Deterioration	Every waiting patient has an acuity-linked reassessment schedule. Reported change or overdue vitals clears prior sign-off and triggers re-triage.
Operational forecast	A separate flow view projects patients in treatment, patients waiting and the staffed-space ceiling. It informs ordering only.

Auditability	Every recommendation, reveal, mismatch, override, reassessment and movement is timestamped with actor and model/ruleset versions.
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Source hierarchy

When sources conflict, engineering must resolve them in the following order:

1. This PRD and its acceptance criteria.
2. The current reference frontend, for interaction states and visual intent.
3. Clinician-approved configuration and deployment policy.
4. The original Round 1 concept, for product principles and architecture lineage.

1. Product context and problem

Emergency triage is often captured as a one-time judgment even though the patient continues to change. Symptoms overlap, presentation varies by age and condition, and some patients under-report. The nurse must make a defensible decision quickly while managing several patients and incomplete records. During a busy shift, even a clinically correct first assessment can become stale.

ATRIA addresses the sequencing problem: who requires attention now, and who is becoming sicker while waiting? It combines nurse judgment, a transparent acuity recommendation, deterministic safety guardrails, repeated observation and operational context. The system is designed to reduce cognitive load by showing one patient, one required decision and one clear next action at a time.

1.1 Round 1 lineage

Round 1 principle	How the current product implements it
Triage is a changing state	Vitals age, reassessment intervals, worsening reports and simulation events continually update the queue.
Near-zero critical undertriage	Layer 0 deterministic gates bypass learned scoring and force critical patients to ESI 1/top of queue.
Dynamic re-ranking	Within-ESI ranking combines wait trajectory, recheck state, uncertainty and

	operational pressure.
Human authority	The nurse assesses first, sees ATRIA second, signs the final ESI and documents clinically significant disagreement.
Decision support, not treatment	ATRIA changes order of attention and recommends acuity; it never prescribes a diagnosis, medicine or procedure.
Missing-data safety	ATRIA abstains when essential vitals are unavailable and requires immediate human assessment.

1.2 Product thesis

North-star behavior

The closer a patient is to critical physiological failure—or the faster their condition is deteriorating—the closer their wait time should move toward zero, without hiding uncertainty or removing nurse authority.

1.3 Problem boundaries

- The product optimizes attention sequencing and reassessment; it is not a diagnostic engine.
- The operational forecast supports flow planning; it must never override a clinical red flag or alter the nurse's signed ESI without new clinical evidence.
- The model must be usable across roughly 100 to 500+ ED visits per day and across mixed technical maturity.
- Approximately half of arriving patients may have a prior record; absence of a record is expected and must degrade safely.

2. Objectives and success measures

2.1 Product objectives

- Reduce critical undertriage and delayed recognition of deterioration.
- Reduce nurse decision fatigue through a blind, one-decision-at-a-time workflow.
- Make every ATRIA output understandable in seconds with short, traceable reasons.
- Keep queue order responsive to new vitals, worsening reports, wait duration and safe operational capacity.

- Operate safely when records, vitals, beds or staff-roster integrations are unavailable.
- Produce a complete, immutable history of clinical recommendations, human sign-offs and overrides.

2.2 Prototype success metrics

Metric	Prototype target	Measurement
Layer 0 critical-case capture	100% on the agreed guardrail test suite	Deterministic unit and integration tests
Essential-vital abstention	100% when HR, BP or SpO ₂ is missing/unparseable	Schema and inference tests
Critical undertriage	Zero in synthetic acceptance suite; clinical target set before pilot	Adjudicated retrospective test set
Queue update latency	≤2 seconds p95 after new observation or flow event	Client and service telemetry
Recommendation latency	≤1 second p95 end to end; model inference ≤250 ms p95	Inference tracing
Blind-assessment integrity	ATRIA ESI and rationale never exposed before nurse selection	UI state tests
Audit completeness	100% of decisions contain actor, patient, timestamps and versions	Audit schema validation
Nurse comprehension	≥90% correctly explain the result and next action in formative testing	Task-based usability study
Tablet task completion	≥95% complete common workflow without zoom or hidden controls	11-inch tablet test

Clinical performance thresholds beyond the synthetic suite must be set by the clinical safety board using representative local data. The prototype targets above are engineering gates, not claims of medical efficacy.

3. Users, roles and jobs to be done

Role	Primary job	Product needs
Triage nurse	Assess, assign ESI, notice deterioration, move patients safely	Fast patient summary; large touch targets; nurse-first workflow; visible anomalies; immediate next action
Charge nurse	Monitor the waiting room and redistribute attention	Queue movement, recheck alerts, surge signal, staffed capacity and override visibility
ED clinician	Receive a prioritized patient and understand why	Signed ESI, latest vitals, reason codes, chronology and data provenance
Operations lead	Match staffing and spaces to near-term demand	Intuitive one-hour forecast, integration status, staffing slider and capacity assumptions
Clinical informaticist	Configure rules and validate safe behavior	Versioned thresholds, test harness, audit export and site-specific settings
ML engineer	Train, deploy and monitor calibrated acuity recommendations	Stable input/output contract, abstention policy, labels, evaluation and versioning
Security/compliance	Protect PHI and demonstrate accountability	Least privilege, encryption, immutable audit events,

		retention and access monitoring
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3.1 Core nurse jobs

- Tell me which patient needs my attention next without making me decode a dense table.
- Let me judge ESI before the system influences me, then show whether we agree.
- Show the few clinical facts that explain the system's view, including missing or age-adjusted data.
- Warn me when a waiting patient needs new vitals or has worsened, and make re-assessment the obvious next action.
- Let me override the system when clinically justified, without hiding the disagreement.
- Do not make me interpret a forecast to know whether capacity is full; state the implication plainly.

3.2 Human-factors principles

Principle	Required design response
One primary decision	Before reveal, the only primary clinical controls are the five ESI choices; other controls are secondary.
Action affordance	Clickable elements use a raised/beveled treatment, hover/focus state and clear label. Informational results use flat shaded cards.
No anchoring	ATRIA remains locked until the nurse selects an ESI.
Progressive disclosure	Show rationale and comparison only after the nurse acts.
Semantic color	Red = concern/critical; amber = review/uncertainty; green = reassuring/match. Never rely on color alone.
Motion with meaning	Animate queue reordering so rank changes

	are perceivable; honor reduced-motion preferences.
No automatic clinical action	Timer expiry creates a prompt and audit event; it never assigns ESI or moves a patient.

4. Scope and release definition

4.1 Full-stack prototype scope

- Synthetic and de-identified patient ingestion; optional FHIR sandbox connectivity.
- Deterministic guardrail service and age-aware vital interpretation.
- Five-class ESI recommendation service with calibrated confidence and abstention.
- Dynamic attention queue with reassessment and deterioration events.
- Nurse-first assessment, comparison, sign-off and override workflow.
- Patient record aggregation with provenance and data-quality state.
- Operations & Flow forecast using staffing, spaces, arrivals and connected-system state.
- Audit and general history views, plus export for validation.
- Offline simulation mode that cannot be confused with production mode.

4.2 Explicitly out of scope for the prototype

- Diagnosis, treatment recommendations, medication dosing or autonomous disposition.
- Replacing hospital emergency procedures, disaster-triage policy or clinician judgment.
- Production claims of ESI accuracy, mortality reduction or regulatory clearance.
- Billing, insurance authorization, appointment scheduling or inpatient order entry.
- Computer vision, ambient audio capture or patient-facing self-triage.
- A fully optimized bed-management algorithm; the forecast is a transparent planning aid.

4.3 Deployment profiles

Profile	Typical daily visits	Minimum integration	Behavior
Small / rural ED	~100–180	Manual intake + optional vitals	Conservative capacity assumptions; fewer staffed spaces; visible

			manual-verification cues
Community ED	~180–400	Records + vitals preferred	Balanced flow forecast and queue updates
Large urban ED	400+	Records, beds, roster and vitals	High-volume event processing, surge state, robust integration monitoring

5. Experience architecture

The product has three primary tabs. Shared counters and flow actions remain in the top ribbon so the nurse can understand shift state without leaving the clinical view.

Surface	Purpose	Primary content
Assessment	Complete the current nurse-first triage decision	Attention queue; selected patient; ESI picker/comparison; latest record; reassessment controls
Operations & Flow	Understand one-hour demand versus staffed capacity	Nurse slider and icons; connected systems; forecast chart; open spaces, wait buffer and arrivals
History	Review accountability and chronology	Audit Log first; General Log second; mismatch/override detail
Top ribbon	Maintain shift context and simulation controls	Checked in; waiting; inside; checked out; current flow; check-in/out actions;

		simulation
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5.1 Assessment layout

Column	Content	Priority
Left	Scrollable waiting queue, attention rank, compact status	Secondary navigation
Center	Nurse assessment and final decision controls	Primary task
Right	Patient record: vitals, reported/observed findings, history and gaps	Clinical evidence

The three columns remain visible together at desktop/tablet landscape widths. The page may scroll vertically, but the queue itself may scroll independently. At narrow widths the layout may stack, with the assessment before the record.

5.2 Interaction states

State	Center panel behavior	Required action
Nurse first	ATRIA locked; ESI picker is raised and prominent	Choose ESI 1–5
Compared—match	Flat nurse and ATRIA result cards; green confirmation	Confirm ESI and continue
Compared—nurse more urgent	Amber escalation message	Confirm; log difference automatically
Compared—nurse less urgent	Amber override message	Enter reason before final sign-off

Critical guardrail	Red alert; ATRIA ESI 1; top-of-queue status remains visible	Confirm/send inside or document override
Diagnostic uncertainty	ATRIA abstains; missing-data reason displayed	Complete human check/vital set; reason if signing off
Recheck due/worsening	Prior sign-off cleared; recheck banner becomes dominant	Retake vitals and repeat ESI

5.3 Accessibility and tablet behavior

- Minimum touch target: 44×44 CSS pixels; primary ESI buttons target ≥56 pixels high.
- All interactive elements require keyboard focus, visible focus outline, accessible name and logical tab order.
- Color states must include text/icon labels such as Red flag, Data missing, Vitals due or ESI match.
- Queue motion uses approximately 650 ms easing and a temporary up/down highlight; reduced-motion mode replaces movement with a short highlight.
- No critical information is available only on hover.
- Text must remain usable at 200% zoom with no loss of actions or data.

6. End-to-end workflows

6.1 New arrival to nurse sign-off

1. Create an ED encounter and patient identity stub; mark source and completeness.
2. Capture or request essential vitals. The red-flag gate runs immediately on every new observation set.
3. Insert the patient into an ESI safety band. If essential vitals are missing, place them in the Diagnostic Uncertainty band between ESI 1 and ESI 2 for immediate human attention.
4. Render the updated attention queue and animate any rank changes.
5. The nurse opens the patient, reviews the record and selects ESI 1–5 without seeing ATRIA's recommendation.
6. ATRIA reveals its recommendation, confidence/abstention state and two to three short rationale bullets.
7. The interface resolves match, nurse escalation, nurse downgrade, guardrail or uncertainty behavior.

8. The nurse confirms, documents an override when required, and either sends the patient inside or advances to the next unassessed patient.
9. The audit service stores recommendation, sign-off, reason, actor, timestamps, data snapshot and model/ruleset versions.

6.2 Waiting-room deterioration

1. A reassessment scheduler calculates the next-due time from signed ESI and site configuration.
2. A due vital set or a staff-reported change marks the patient as needing reassessment and boosts within-band attention.
3. A reported worsening event immediately clears any prior nurse ESI sign-off in the active workflow.
4. New vitals are evaluated by Layer 0 before any model. A new red flag forces ESI 1/top of queue.
5. If the ESI recommendation changes, the queue moves visibly and the nurse repeats blind sign-off.
6. The History tab records both the deterioration signal and the resulting decision.

6.3 Flow planning

1. Aggregate current waiting, inside, checked-out, arrival-rate, nurse-count and staffed-space data.
2. Project five points: now, +15, +30, +45 and +60 minutes.
3. Display green patients-in-treatment, orange patients-waiting and a dashed staffed-space ceiling.
4. Cap the treatment line at staffed spaces. The waiting line may rise when demand exceeds safe intake; this is not a claim that beds exceed capacity.
5. Translate the forecast into one sentence and three metrics: open staffed spaces, median wait buffer and next-hour arrivals.
6. Apply any operational boost only within the same ESI safety band. Log the configuration that influenced ranking.

7. Functional requirements — Assessment

ID	Requirement	Acceptance evidence
ASS-001	Show the selected patient's name, age, sex, wait, patient/encounter identifier and main concern before	Fields render from a versioned patient snapshot; missing values are labeled,

	the ESI decision.	never blank.
ASS-002	Keep ATRIA's ESI, confidence and rationale hidden until the nurse chooses an ESI.	Automated UI test confirms no recommendation data is present in the accessible DOM before selection.
ASS-003	Present ESI as five visibly clickable, touch-friendly controls using the approved nomenclature.	Each control has number, name, hover/focus/pressed state and ≥ 44 px target.
ASS-004	After selection, replace the picker emphasis with flat Nurse ESI and ATRIA Recommendation result cards.	Result cards display full ESI label and do not look clickable.
ASS-005	Generate one outcome state: match, nurse escalation, nurse downgrade, critical guardrail or diagnostic uncertainty.	Exactly one outcome component and one primary next action appear.
ASS-006	Allow the nurse to change ESI before final sign-off.	Change ESI returns to blind selection and does not leak the previous ATRIA recommendation.
ASS-007	Require a reason for a less-urgent nurse ESI than ATRIA, any red-flag override, or sign-off while ATRIA abstains.	Final action remains blocked until a non-empty reason is submitted.
ASS-008	Do not require a reason when the nurse selects a more urgent ESI, but record the difference.	Audit event contains nurse and ATRIA ESI plus actor/version.

ASS-009	For final ESI 1 or 2, label the primary action Confirm & send inside; for ESI 3–5, confirm and advance.	Waiting/inside counters and encounter state update atomically.
ASS-010	Display an adaptive Harvey-ball decision cue without numeric countdown or automatic decision.	Progress changes visually; expiry creates a prompt/history event only.
ASS-011	Show two to three reason-code-derived rationale bullets after reveal.	Reasons are traceable to inputs/rules/model contribution; no diagnosis is generated.
ASS-012	Clearly label the workstation as decision support and state that ATRIA changes attention order, not treatment.	Statement is visible after comparison and in product documentation.

7.1 Harvey decision cue

Input	Prototype behavior
Steady load	90 seconds baseline
Busy load	105 seconds baseline
Surge load	120 seconds baseline
High-acuity patient (prototype level ≤ 2)	Subtract 15 seconds
Age <12 or ≥ 75	Add 15 seconds
Bounds	75–135 seconds

Expiry	Prompt nurse; create event; never auto-assign ESI
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These timings are human-factors prototype defaults, not clinical standards. Production values must be configurable by site and validated through usability testing.

8. Functional requirements — Queue, record and history

8.1 Attention queue

ID	Requirement	Acceptance evidence
QUE-001	Display live attention rank, patient name, main concern and one compact status.	Queue remains scannable; no row requires horizontal scrolling.
QUE-002	Re-rank after every relevant clinical or operational event.	New vitals, wait increment, recheck, worsening, staffing, systems, check-in and send-inside all trigger recomputation.
QUE-003	Animate rank changes with a FLIP-style transition and temporary direction highlight.	Nurse can see who moved; reduced-motion preference is supported.
QUE-004	Keep critical guardrail patients above all others and Diagnostic Uncertainty above ESI 2–5.	Ordering tests pass for all pairwise safety-band combinations.
QUE-005	Never let operational forecasting move a patient across an ESI safety band.	Invariant test proves rank score boost is lower than band separation.

QUE-006	Use a scrollable queue independent of the assessment and record panels.	Long queues do not push required clinical actions off screen.
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8.2 Patient record

ID	Requirement	Acceptance evidence
REC-001	Show the most recent HR, BP, SpO ₂ , temperature, respiratory rate and onset with observed time/source.	Every value includes provenance in the data object and accessible detail.
REC-002	Use red, amber and green text/icon states without tinted data-cell backgrounds.	Color conveys severity while the panel remains visually quiet; text labels are available to screen readers.
REC-003	Apply age-aware vital interpretation.	Pediatric control and older-adult comparison cases render different states from identical HR/BP values.
REC-004	Separate reported symptoms from observed findings.	No source ambiguity in UI or data contract.
REC-005	Show conditions, medicines, allergies and last contact only when available; otherwise label Not available or system off.	Missing and disconnected states are distinguishable.
REC-006	Show one consolidated missing-information prompt; do not show a	Current patient record ends with record check/missing information only.

	next-patient block.	
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8.3 History

ID	Requirement	Acceptance evidence
HIS-001	Default History to Audit Log, with General Log as the second menu.	Audit Log is selected on first entry.
HIS-002	Audit Log includes nurse–ATRIA differences and documented manual overrides.	Mismatches are filterable and show nurse ESI, ATRIA ESI, reason, actor and time.
HIS-003	General Log includes assessment, reassessment, queue movement, simulation and audit events chronologically.	All event types render consistently and are time ordered.
HIS-004	History must be append-only for ordinary users.	No edit/delete control; correction creates a new linked event.
HIS-005	Persist model, ruleset and data-snapshot identifiers.	An auditor can reconstruct the decision context.

9. Four-layer clinical safety architecture

Implementation rule

The learned model must never be able to suppress Layer 0. Missing essential data must never fall through to a normal prediction. Layer outputs and reason codes are stored separately so the final result remains reconstructable.

Layer	Responsibility	Output	Authority
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0 — Red-flag gate	Deterministic immediate-physiology and documented-red-flag rules	Critical / uncertain / continue + reason codes	Can force ESI 1 or abstention; cannot be overridden silently
1 — Acuity recommender	Predict ESI 1–5 from the validated clinical snapshot	Calibrated probabilities, recommended ESI, confidence, explanation inputs	Advisory; hidden until nurse choice
2 — Dynamic re-ranker	Order attention within safety bands using time, recheck, uncertainty and flow	Rank score, rank, movement reason	May change order, not treatment or signed ESI
3 — Human authority	Nurse assessment, sign-off, escalation, override and disposition	Final ESI, action and documented reason	Final clinical decision; disagreement remains visible/audited

9.1 Layer 0 deterministic rules

ID	Requirement	Acceptance evidence
SAF-001	Evaluate Layer 0 before model inference and again after every new vital or documented finding.	Rule trace precedes inference trace; critical cases can skip model serving.
SAF-002	SpO ₂ <90% triggers critical.	Boundary tests at 89.9/90.0.
SAF-003	Age <15 uses minimum SBP = 70 + 2×age; age ≥15 uses minimum SBP 90 mmHg.	Boundary tests for ages 3, 14, 15 and 75.

SAF-004	Adult respiratory rate <10 or >30/min triggers critical.	Boundary tests at 9/10/30/31.
SAF-005	Documented GCS ≤ 8 , active seizure, airway compromise or uncontrolled hemorrhage triggers critical.	Structured finding codes preferred; text extraction is not the sole production mechanism.
SAF-006	Critical status forces ATRIA ESI 1 and top-of-queue position.	No heuristic/model output can reduce urgency.
SAF-007	Missing/unparseable HR, BP or SpO ₂ triggers Diagnostic Uncertainty and model abstention.	No normal prediction or green tone is produced.
SAF-008	A red-flag override requires a reason and retains the active critical banner.	Override event is immutable and immediately visible to charge nurse.

9.2 Age-aware tones

Vital	Bad/critical presentation	Warning	Good
SpO ₂	<90%	90–93%	$\geq 94\%$
SBP	Below age minimum or >180	Within 10 mmHg of minimum or >160	Otherwise in prototype range
HR	Outside age-specific range	Within 10 bpm of either bound	Within central range
Temperature	<35°C or $\geq 39.5^\circ\text{C}$	35–35.9°C or >38.5°C	36–38.5°C

Respiratory rate	Outside age-specific range	Within 2 of lower or 4 of upper bound	Within central range
Missing value	Not classified as normal	Missing/manual verification	Not applicable

Prototype heart-rate ranges: age <5: 70–140; 5–7: 65–130; 8–11: 60–120; 12–14: 55–115; age ≥15: 50–110. Prototype respiratory ranges: age <5: 18–34; 5–11: 14–30; 12–14: 12–26; adult: 10–30. These must be externalized and clinically approved.

9.3 Missing-data and ambiguity protocol

- Essential missing data is a system state, not a weak negative signal.
- ATRIA returns status=ABSTAIN, recommended_esi=null, confidence=null and explicit missing fields.
- The patient is placed below active ESI 1 red flags and above scored ESI 2–5 patients until human assessment resolves the gap.
- The UI shows Complete vital set now and keeps the patient in the recheck workflow.
- After a complete set arrives, Layer 0 and the model run on a new immutable snapshot; the nurse repeats ESI selection.

10. ESI decision product specification

ESI is a five-level triage framework. In ATRIA, ESI is a clinical acuity and resource-needs classification; the attention rank is a separate, continuously changing sequence. The UI must never use rank as a synonym for ESI.

Level	Nomenclature	Product interpretation
ESI 1	Resuscitation	Immediate life-saving intervention / deterministic critical guardrail
ESI 2	Emergent	High risk, severe symptoms or time-sensitive concern requiring rapid attention
ESI 3	Urgent	Stable enough to wait briefly; likely multiple resources

ESI 4	Less urgent	Stable; likely one resource
ESI 5	Non-urgent	Stable; likely no resources beyond examination

AHRQ describes ESI as a five-level ED triage algorithm stratifying patients from 1 (most urgent) to 5 (least urgent) based on acuity and resource needs. Local implementation must follow the hospital's licensed/approved ESI training and governance; the descriptions above are product-level labels, not a replacement algorithm.

10.1 Nurse–ATRIA comparison matrix

Case	Definition	UI	Audit/action
Match	Nurse ESI = ATRIA ESI	Green ESI match	Confirm; normal event
Nurse escalation	Nurse ESI is numerically lower/more urgent	Amber; human concern moves forward	No reason required; mismatch logged
Nurse downgrade	Nurse ESI is numerically higher/less urgent	Amber override warning	Reason required before final action
Guardrail	Layer 0 critical	Red persistent critical banner	ESI 1 expected; any lower urgency requires reason/escalation
Uncertain	Essential data missing	Amber abstention	Human check; reason required if finalizing before resolution

10.2 Recommendation response

The recommendation service must return the following conceptual payload. Exact transport

naming may vary, but semantics may not.

```
JSON
{
  "assessment_id": "asm_01J...",
  "patient_id": "PT-1042",
  "snapshot_id": "snap_01J...",
  "status": "RECOMMEND | CRITICAL | ABSTAIN",
  "recommended_esi": 1,
  "probabilities": {"1": 0.97, "2": 0.02, "3": 0.01, "4": 0, "5": 0},
  "confidence": 0.97,
  "reason_codes": ["SPO2_BELOW_90", "SBP_BELOW_AGE_MIN"],
  "rationale": ["SpO2 87% is below the 90% guardrail",
    "Age-adjusted SBP triggered: 82 < 90 mmHg"],
  "ruleset_version": "guardrail-us-ed-1.0.0",
  "model_version": "esi-prototype-0.1.0",
  "created_at": "2026-08-27T14:12:04Z"
}
```

11. Dynamic attention-ranking specification

11.1 Safety-band ordering

Order	Band	Prototype numeric base	Invariant
1	Layer 0 critical	1,000,000 + wait	Always above every other band
2	Diagnostic uncertainty	45,000 + capped wait	Below critical; above scored ESI 2–5
3	ESI 2	40,000 + modifiers	Never crossed by ESI 3 modifiers

4	ESI 3	30,000 + modifiers	Never crossed by ESI 4 modifiers
5	ESI 4	20,000 + modifiers	Never crossed by ESI 5 modifiers
6	ESI 5	10,000 + modifiers	Lowest scored acuity band

The numeric bases are an implementation technique, not clinical scores. Production code should represent the band explicitly and sort lexicographically so future modifiers cannot accidentally cross safety boundaries.

11.2 Within-band modifiers

Modifier	Prototype default	Purpose
Sepsis trajectory	Temp >38.5°C and age-aware tachycardia: (wait_minutes/60)×5	Increase attention progressively while waiting
Five-hour safeguard	If not sepsis trajectory and wait ≥300 min: +50	Prevent extremely long waits from being ignored
Worsening reported	+140	Move patient toward immediate reassessment within band
Vitals recheck due	+55	Surface stale assessment
Record unavailable	+12 when record incomplete	Add caution for uncertainty
Vitals connection unavailable	+8	Add caution/manual-verification state
Known information gap	+4 connected / +10	Reflect incomplete context

	disconnected	
Operational outlook	0–18 based on shortage/load and wait	Prioritize older waits among clinically similar patients
Tie-break	Longer wait, then earlier arrival, then stable patient ID	Deterministic ordering

11.3 Rank explanation

- Store a structured reason list for every rank computation; do not try to infer an explanation later from the final number.
- Expose only the top two or three reasons in the nurse UI; retain the full trace in the audit record.
- A queue-movement event includes previous rank, new rank, ESI band, triggering event and timestamp.
- Operational reasons must be worded as planning signals, never as clinical deterioration.

11.4 Sorting pseudocode

Plaintext

for each patient:

 layer0 = evaluate_guardrails(latest_snapshot)

 if layer0.critical: band = CRITICAL

 else if layer0.uncertain: band = DIAGNOSTIC_UNCERTAINTY

 else: band = ATRIA_OR_SIGNED_ESI

 within_band = time_trajectory + recheck + data_uncertainty + flow_modifier

sort by:

 band urgency ascending,

 within_band score descending,

 arrival timestamp ascending,

 patient id ascending

12. Reassessment and deterioration

ID	Requirement	Acceptance evidence
REA-001	Assign a next-reassessment timestamp whenever an ESI is signed or vitals are captured.	Scheduler persists due_at and interval version.
REA-002	Prototype defaults: ESI 1=5m, ESI 2=15m, ESI 3=45m, ESI 4=90m, ESI 5=180m.	Intervals are configuration, not hard-coded UI constants.
REA-003	When due, set recheck_due=true, create alert and re-rank within the band.	Queue tag and record banner update within 2 seconds.
REA-004	A nurse can report change at any time.	Action marks worsening, requests immediate vitals and clears prior active sign-off.
REA-005	New observations create a new snapshot and rerun Layer 0 and Layer 1.	Old snapshot remains immutable; recommendation is linked to new snapshot.
REA-006	If new recommendation changes ESI, require a new blind nurse sign-off.	Previous ATRIA output is not shown until the new nurse choice.
REA-007	Do not silently downgrade because time passed or flow improved.	ESI changes only from new clinical assessment; rank may still change.

REA-008	Escalate overdue reassessment to charge nurse based on configurable grace periods.	Escalation notification and acknowledgement are audited.
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12.1 Reassessment state machine

Current state	Event	Next state	System action
Current	Due time reached	Vitals due	Alert, queue modifier, history event
Any	Nurse reports change	Worsening	Clear sign-off; immediate vitals request
Vitals due/worsening	New complete vitals	Re-assessment required	Run guards/model; refresh queue
Re-assessment required	Nurse selects ESI	Compared	Reveal ATRIA and resolve outcome
Compared	Final sign-off	Current	Schedule next check; persist audit
Any	Layer 0 critical	Critical	Force ESI 1/top queue and alert

13. Operations & Flow

Operations & Flow is separated from the clinical assessment to reduce cognitive overload. It explains demand versus staffed capacity in one glance and contributes only bounded within-ESI ranking context.

13.1 Controls and connected systems

Input	Effect	Degraded-mode behavior
Nurses available	Changes staffing capacity, flow state, nurse icons and projection	Manual entry remains available; source is recorded
Records	Improves history completeness and model certainty	Gaps labeled; caution modifier; no invented history
Beds/spaces	Provides physical/treatment-space ceiling	Apply conservative reduction to assumed spaces
Roster	Validates actual staff coverage	Apply staffing safety margin
Vitals	Provides live observation feed	Manual verification; missing essentials cause abstention

13.2 Flow state

The prototype calculates a capacity ratio from waiting patients, available nurses and connected-system efficiency. It labels the shift Steady, Busy or Surge. Production thresholds must be site-configurable and should include recent arrival rate, triage throughput, staffed treatment spaces and data freshness.

ID	Requirement	Acceptance evidence
OPS-001	Show nurses available as a number, slider and matching nurse icons.	Changing the slider immediately updates icons, flow state and forecast.
OPS-002	Show connected-system status and the specific effect of any disconnection.	Each toggle changes both visible copy and computation assumptions.
OPS-003	Project in-treatment and waiting counts at 15-minute	Values are reproducible from the forecast input

	intervals for one hour.	snapshot.
OPS-004	Display a horizontal staffed-space ceiling; never imply physical beds grow with demand.	Treatment projection is capped; waiting may exceed capacity.
OPS-005	Explain the graph in one sentence that states the operational consequence.	Comprehension test: user identifies whether queue grows and whether spaces fill.
OPS-006	Display open staffed spaces, median wait buffer and expected next-hour arrivals.	Definitions are available and values update with inputs.
OPS-007	Keep operational modifier bounded within ESI.	Invariant tests show no cross-band reorder.
OPS-008	Store forecast version, inputs and generated points.	Historical queue decisions are reproducible.

13.3 Forecast definitions

Visual/metric	Meaning
Green line — In treatment	Projected number occupying a staffed treatment space; always capped at staffed capacity.
Orange line — Waiting	Projected patients still in the waiting queue; can rise even when all staffed spaces are full.
Dashed red line — Staffed spaces	Current care spaces that are both physically available and safely staffed; not

	total licensed beds.
Open staffed spaces	$\max(0, \text{staffed_space_limit} - \text{patients_inside})$.
Wait buffer	Median remaining time to site-configured wait threshold among non-critical waiting patients; planning signal only.
Arrivals	Expected arrivals over the next hour from recent/site arrival rate.
Decision pattern	Recent proportion of signed patients sent inside; used only in transparent forecast scenarios.

14. ML model product specification

Recommended prototype strategy

Build a hybrid system: deterministic Layer 0 + a calibrated, interpretable tabular/ordinal ESI model + deterministic queue bands. Do not put a generative model or free-text explanation generator in the safety-critical path.

14.1 Problem formulation

Element	Specification
Task	Five-class ordinal recommendation: ESI 1, 2, 3, 4 or 5.
Precondition	Layer 0 returned continue and essential data passed completeness/quality checks.
Label	Clinically adjudicated ESI at initial assessment/reassessment; not raw model-influenced nurse choice alone.
Output	Class probabilities, recommended class,

	calibrated confidence, abstention status, reason-code inputs and version.
Objective	Minimize ordinal error with asymmetric penalty for undertriage, especially true ESI 1–2 predicted as less urgent.
Model family	Model-agnostic contract. Start with interpretable gradient-boosted trees or ordinal model; benchmark against rules and simple baselines.
Text	Use structured complaint codes and clinician-entered findings first. If text is used, it requires separate validation, provenance and failure handling.
Decision role	Advisory; hidden until nurse ESI selection; never automatically signs off or orders treatment.

14.2 Minimum feature groups

Group	Examples	Rules
Demographics	Age, age band, sex if clinically justified	Age required for thresholds; do not use race/ethnicity as a predictive shortcut.
Arrival	Mode, timestamp, wait, referral source	No post-disposition leakage.
Complaint	Structured complaint, onset, reported severity	Retain source and capture time.
Vitals	HR, SBP/DBP, SpO ₂ , RR, temperature, glucose if	Latest value plus trend, age and device/source;

	present	missingness explicit.
Neurological/airway	GCS, seizure, airway compromise	Layer 0 structured inputs; never rely only on NLP.
History	Conditions, medicines, allergies, pregnancy	Use only available/current data; include availability indicator.
Trajectory	Change from prior vitals, vital age, worsening report	Supports reassessment recommendation; no future data.
Resource context	Expected resources if part of local ESI protocol	Clinical model only; capacity is excluded from ESI prediction.

14.3 Features prohibited from the clinical ESI model

- Current bed availability, nurse count, current flow state or hospital revenue.
- The nurse's current ESI selection before the model generates its independent recommendation.
- Post-triage treatment, admission, length of stay or future outcome data at inference time.
- Protected attributes without an approved clinical rationale and fairness review.
- Free-text embeddings or external data whose provenance, version and latency cannot be audited.

14.4 Confidence and abstention

ID	Requirement	Acceptance evidence
ML-001	Calibrate probabilities on held-out temporal/site data.	Reliability curves and calibration error reported overall and by subgroup.
ML-002	Abstain for essential missingness before model inference.	Deterministic missing-data tests pass.

ML-003	Support low-confidence abstention using a clinically approved threshold.	Prototype default may be 0.70 max probability; threshold is configurable and validated.
ML-004	Abstain on out-of-distribution inputs or incompatible schema/model versions.	OOD/schema tests return explicit status, never a default ESI.
ML-005	Do not use confidence to soften a Layer 0 critical result.	Critical output contains guardrail reasons regardless of model.
ML-006	Return stable reason codes and top contributing clinical factors.	Rationale text is templated from audited factors.

14.5 Explainability

- System Rationale contains two or three short bullets, ordered by clinical importance.
- Layer 0 bullets quote the observed value and threshold, e.g., Age-adjusted SBP triggered: 82 is below 90 mmHg.
- Model bullets describe factors, not causality: e.g., high-risk complaint pattern, abnormal age-adjusted heart rate, incomplete record.
- Queue bullets are separate from clinical rationale: e.g., reassessment due or longer wait within ESI 3.
- The audit trace retains all feature values, transformations, missingness flags, probabilities and reason codes permitted by policy.

15. Data model and contracts

15.1 Core entities

Entity	Required fields	Notes
Patient	patient_id, age/date of birth, sex, identity status	Separate identifiers from clinical snapshots; support

		unknown identity.
Encounter	encounter_id, patient_id, arrival_at, status, location	Status: arrived, waiting, in-treatment, completed.
ClinicalSnapshot	snapshot_id, encounter_id, captured_at, source refs, structured findings	Immutable input bundle for one guardrail/model decision.
VitalObservation	code, value, unit, observed_at, source, quality	Use standardized codes/units; retain raw and normalized values.
AtriaAssessment	assessment_id, status, ESI probabilities/recommendation, reasons, versions	Linked to exactly one snapshot.
NurseAssessment	nurse_assessment_id, ESI, actor, selected_at, final_at	Independent selection timestamp precedes reveal.
QueueEntry	encounter_id, safety_band, score, rank, reasons, calculated_at	Current state plus movement-event history.
ReassessmentTask	due_at, interval, state, trigger, acknowledged_by	Config/version required.
OverrideEvent	nurse ESI, ATRIA ESI/status, reason, actor, time	Append-only; links to assessment and snapshot.
OperationalSnapshot	waiting, inside, nurses, spaces, arrival rate, systems	Input to forecast and flow label.
Forecast	horizon, points, staffed ceiling, assumptions, version	Reproducible; never modifies ESI.

15.2 Essential clinical fields

Field	Type/unit	Required for model	Validation/failure
age	integer years + DOB if known	Yes	Non-negative; pediatric rule uses age <15
heart_rate	number bpm	Yes	Missing/unparseable → abstain
blood_pressure	SBP and DBP mmHg	Yes	SBP required; missing/unparseable → abstain
spo2	number %	Yes	Missing → abstain; <90 → critical
respiratory_rate	number/min	Strongly expected	Adult <10/>30 → critical when present; missing is explicit
temperature	number °C	Expected	Unit normalize; missing explicit
complaint	coded concept + text	Yes	Unknown allowed; must not invent code
gcs	integer 3–15	When assessed	≤8 → critical
observed_at	UTC timestamp	Yes	Reject future/stale beyond configured limit
source	device/manual/EHR	Yes	Disconnected/unverified sources labeled

15.3 Provenance and time

- All timestamps are stored in UTC and rendered in local hospital time with timezone.
- Every normalized observation retains the raw value, unit, source system, source identifier and normalization version.
- A model decision references an immutable snapshot; later corrections create a new snapshot and superseding assessment.
- Data freshness is computed from observed_at, not interface arrival time.

15.4 FHIR-oriented mapping

ATRIA concept	FHIR R4 mapping
Patient identity/demographics	Patient
ED arrival and status	Encounter
Vitals and point-in-time findings	Observation
Problem list/history	Condition
Medicines	MedicationStatement and/or MedicationRequest, per source
Allergies	AllergyIntolerance
Clinical task/reassessment	Task
Assessment output	RiskAssessment or GuidanceResponse only after integration-governance review; retain native ATRIA record regardless

16. Service and API contract

16.1 Recommended service boundaries

Service	Responsibility
Clinical data gateway	Validate/normalize patient, encounter, vital and record data; maintain provenance.
Guardrail service	Run deterministic Layer 0 rules from a signed/versioned configuration.
Inference service	Serve the ESI model, calibration and model reason codes.
Assessment orchestrator	Create snapshots; sequence guardrail → model; enforce blind reveal state.
Queue service	Compute safety band, within-band modifiers, ranks and movement events.
Reassessment scheduler	Create due tasks, escalation and acknowledgement events.
Flow forecast service	Project waiting/treatment against staffed spaces; return explanation.
Audit service	Append-only decision and access event store; export and retention controls.
Integration adapters	FHIR/HL7 records, vital devices, beds and staff rosters; health status.

16.2 Endpoint surface

Method/path	Purpose	Key response
POST /v1/encounters	Check in patient	Encounter + initial queue entry
POST /v1/encounters/{id}/observat	Add/retake vitals	Snapshot, guardrail state,

ions		reassessment requirement
POST /v1/encounters/{id}/nurse-assessments	Store blind nurse ESI	Nurse assessment; reveal token
POST /v1/assessments/{id}/reveal	Reveal ATRIA after nurse selection	Recommendation, comparison state and rationale
POST /v1/assessments/{id}/finalize	Final nurse sign-off/action	Encounter state, queue update, audit id
POST /v1/assessments/{id}/override	Document required disagreement	Immutable override id
POST /v1/encounters/{id}/worsening	Report change	Reassessment task and queue update
GET /v1/queue	Read current ranked attention list	Versioned queue snapshot
GET /v1/operations/forecast	Read one-hour forecast	Points, ceiling, metrics, explanation
PATCH /v1/operations/staffing	Update available nurses	Operational snapshot and recompute id
GET /v1/history?mode=audit general	Read history	Paginated immutable events
GET /v1/integrations/health	Read connected-system status	Freshness, latency and failure state

16.3 Real-time events

Event	Required payload
queue.updated	queue_version, changed encounter ids, old/new ranks, reason codes
assessment.recommendation_ready	assessment id and reveal eligibility; never pre-reveal recommendation data
patient.reassessment_due	encounter, due time, ESI, escalation level
patient.worsening_reported	encounter, reporter, time, structured note
patient.guardrail_triggered	encounter, reasons, values, threshold/ruleset
operations.snapshot_updated	counts, nurses, spaces, systems and flow state
audit.override_created	override id and actor; restricted payload by role

Use server-sent events or WebSockets for live updates. Every event must be idempotent, ordered by a monotonically increasing stream version and recoverable by fetching the latest snapshot after a disconnect.

17. Integration and degraded-mode requirements

ID	Requirement	Acceptance evidence
INT-001	Expose health, freshness and last-success time for Records, Beds, Roster and Vitals.	A stale connection is distinguishable from a healthy one.
INT-002	Records disconnection	History shows unavailable;

	must not block triage.	model uses availability flags or abstains if required.
INT-003	Vitals disconnection requires manual verification and never defaults values.	Missing essentials trigger Diagnostic Uncertainty.
INT-004	Beds disconnection reduces forecast capacity conservatively.	Assumption and margin appear in forecast explanation/audit.
INT-005	Roster disconnection reduces effective staffed capacity.	Manual nurse number remains available with actor/source.
INT-006	Integration recovery creates a new operational/clinical snapshot and recomputation.	No silent replacement of prior evidence.
INT-007	Map data to standardized terminology and units at the gateway.	Raw values retained; normalization failures quarantined.
INT-008	Do not allow simulation data to enter production integration streams.	Environment boundary and synthetic-data labels are enforced.

17.1 Data maturity matrix

Maturity	Available systems	Permitted prototype behavior
Low	Manual intake only	Rules/model run only on manually verified values; conservative capacity; explicit uncertainty

Medium	Records + some vitals	Aggregate history; automatic vital updates where trusted; manual staff/spaces
High	Records + beds + roster + vitals	Near-real-time reassessment, staffed-space forecast and richer provenance

18. Security, privacy, audit and regulatory assumptions

18.1 Jurisdiction

The prototype assumes deployment in the United States and a HIPAA-oriented operating model. The interface currently mentions HIPAA/GDPR; production copy must be jurisdiction-specific and approved by legal/compliance. A product must not claim compliance merely because it records an override.

18.2 Security requirements

ID	Requirement	Acceptance evidence
SEC-001	Encrypt PHI in transit and at rest using organization-approved standards.	Security architecture and penetration evidence.
SEC-002	Use role-based access with least privilege for nurse, charge nurse, clinician, operations, administrator and auditor.	Authorization tests for every endpoint and event.
SEC-003	Require authenticated actor identity for sign-off, override and operational	No anonymous clinical mutation.

	changes.	
SEC-004	Maintain append-only audit controls for create/read/update/export activity involving ePHI and clinical decisions.	Audit events are tamper-evident and queryable.
SEC-005	Apply minimum-necessary display and export.	Operations users do not receive unnecessary clinical detail.
SEC-006	Protect availability with tested backups, downtime procedure and recovery objectives.	Restore drill and degraded-mode test.
SEC-007	Define configurable retention and legal-hold policy before real data use.	Approved policy linked to storage lifecycle.
SEC-008	Separate synthetic, test and production tenants and keys.	No cross-environment identifiers or data flows.

18.3 Override audit record

Field group	Required content
Identity	override_id, patient_id, encounter_id, assessment_id, snapshot_id
Decision	nurse ESI, ATRIA status/ESI/probabilities, final action
Reason	selected reason code + free-text clinical justification

Actor	user id, role, organization/location, authenticated session id
Time	nurse selection, reveal, reason entry, finalization timestamps
Versions	UI build, API, ruleset, model, calibration, terminology and configuration versions
Evidence	hash/reference to immutable input snapshot and relevant observations
Integrity	event sequence, prior-event hash/signature as approved by security design

18.4 Regulatory boundary

Required review before real-world use

The implementation team must obtain regulatory counsel and clinical-safety review to determine whether the intended use and functionality constitute regulated device software. The 2026 FDA Clinical Decision Support Software guidance should be reviewed against the exact claims, transparency, automation and user population.

19. Non-functional requirements

Area	Requirement
Performance	Recommendation ≤ 1 s p95 end to end; queue update ≤ 2 s p95; history first page ≤ 2 s p95.
Availability	Prototype target 99.5% during test windows; production SLO set by deployment risk assessment.
Consistency	Finalization and counter/queue movement are atomic; duplicate requests are idempotent.

Scale	Support at least 500 visits/day, 200 concurrent waiting encounters and 50 concurrent clients per site in prototype load test.
Offline/degraded	Read last trusted snapshot; permit policy-approved manual workflow; clearly label stale/offline state.
Accessibility	Target WCAG 2.2 AA; keyboard, screen reader, contrast, zoom and reduced motion.
Browser/tablet	Current Chrome/Edge/Safari and 11-inch landscape tablet; responsive fallback for smaller screens.
Observability	Distributed trace from data ingestion through rules/model/queue; PHI-safe logs and metrics.
Maintainability	Rules, thresholds and intervals live in signed versioned configuration; no production magic numbers in UI code.
Testability	Deterministic synthetic fixture pack; replayable event stream; frozen model artifacts.

19.1 Failure behavior

Failure	Required behavior
Model unavailable	Layer 0 continues; ATRIA shows Recommendation unavailable; nurse can proceed with audited manual sign-off.
Guardrail service unavailable	Fail closed for ATRIA recommendation;

	show safety-service unavailable; invoke downtime procedure.
Queue service unavailable	Freeze last known ordering, label stale time and use hospital manual triage queue.
Audit store unavailable	Do not finalize an override without durable local/queued write permitted by approved downtime design.
Integration stale	Display stale badge and source age; apply degraded assumptions.
Client disconnect	Reconnect using stream cursor; fetch authoritative state before accepting action.

20. Model development, validation and monitoring

20.1 Dataset requirements

- Use multi-site, demographically representative ED data with sufficient pediatric, geriatric, pregnancy and rare critical cases.
- Create adjudicated labels through a documented clinician-review protocol. Retain raw nurse ESI, adjudicated ESI and outcome separately.
- Split data by patient and time; include at least one site-held-out evaluation to estimate transportability.
- Prevent leakage from post-triage tests, treatments, disposition or outcomes not available at inference time.
- Document missingness and integration availability as part of the dataset, rather than imputing silently.
- Maintain a data sheet covering provenance, inclusion/exclusion, label process, class balance and known limitations.

20.2 Evaluation matrix

Metric	Why it matters	Reporting
--------	----------------	-----------

ESI 1 recall/sensitivity	Critical undertriage is the highest-severity failure	Overall and by site/subgroup
ESI 2 recall	Captures time-sensitive emergent cases	Overall and by site/subgroup
Critical undertriage rate	True ESI 1/2 predicted 3–5	Count, rate and reviewed case list
Overtriage rate	Safety/capacity trade-off	By predicted class and site
Ordinal MAE / weighted kappa	Overall five-level agreement	Confidence intervals
Calibration	Confidence must mean what it says	Reliability curve, ECE/Brier
Abstention coverage/risk	Safe handling of uncertain inputs	Error rate at each coverage level
Queue time-to-attention	Product outcome for waiting-room sequence	Simulation and prospective shadow mode
Subgroup parity	Detect performance differences	Age bands, sex, language/site and approved fairness groups

20.3 Release gates

- 100% deterministic guardrail and missing-data acceptance tests.
- Clinical sign-off on ruleset, ESI nomenclature, reassessment policy and explanation templates.
- Predefined critical-undertriage and calibration thresholds passed on frozen test and external/site-held-out data.
- No subgroup exceeds the approved harm threshold without documented mitigation.
- Shadow-mode review confirms input mappings, latency, alert burden and queue stability.
- Rollback path and prior model/ruleset remain deployable.

20.4 Production monitoring

Signal	Alert/response
Input drift	Compare feature distributions and missingness by site; investigate mapping changes.
Prediction drift	Track class distribution, confidence and abstention.
Outcome/label drift	When adjudicated labels arrive, monitor undertriage and calibration.
Override rate	Review mismatches by direction, clinician, site and model version; do not penalize appropriate overrides.
Recheck compliance	Track overdue tasks and time from worsening report to new vitals.
Alert burden	Track alerts per nurse/shift, acknowledgement and ignored/duplicate alerts.
Integration quality	Freshness, parsing failures, unit mismatches and downtime.
Queue churn	Excessive rank movement may increase cognitive load; monitor moves/patient/hour.

21. Acceptance scenarios

Scenario	Given / When	Then
Pediatric control	Age 3, HR 130, BP 85/52, SpO ₂ 98	No adult-threshold red flag; pediatric thresholds appear in rationale.

Older adult same vitals	Age 75, HR 130, BP 85/52, SpO ₂ 98	Age-adjusted adult SBP breach forces ESI 1/top queue; cardiac/perfusion concern visible.
Missing essential data	John Doe has null HR, BP and SpO ₂	ATRIA abstains; Diagnostic Uncertainty; complete vital set now; no normal score.
Low oxygen	Any age with SpO ₂ 89%	Critical guardrail and ESI 1 regardless of model.
Adult RR boundary	Adult RR 31/min	Critical; 30/min is not triggered by this rule.
Sepsis trajectory	Fever >38.5 plus age-aware tachycardia; wait 180m	Time trajectory increases within ESI band and rationale names it.
Five-hour safeguard	Non-sepsis case reaches 300m	Flat wait safeguard added within band; queue movement logged.
Blind assessment	Nurse has not selected ESI	ATRIA value/probabilities/rationale absent from view and accessible DOM.
Match	Nurse ESI equals ATRIA	One confirmation action; no duplicated send-inside control.
Nurse escalation	Nurse selects ESI 2; ATRIA recommends ESI 3	Nurse view retained; mismatch logged; no mandatory reason.
Nurse downgrade	Nurse selects ESI 3; ATRIA recommends ESI 2	Finalization blocked until reason is saved.

Guardrail override	Layer 0 critical; nurse selects ESI 2–5	Critical banner persists; reason mandatory; charge-nurse escalation event.
Vitals due	Reassessment due time passes	Queue tag/banner appears and within-band priority rises.
Worsening while waiting	Nurse presses Report change	Prior sign-off clears; new vitals required; queue reorders with animation.
Capacity full	In treatment reaches staffed ceiling while arrivals continue	Green line stops at ceiling; orange waiting line may rise; plain-language warning.
Roster disconnected	Roster toggle off	Capacity uses safety margin; explanation and forecast inputs record degraded state.
Model unavailable	Inference times out	Layer 0 remains; ATRIA unavailable; nurse can proceed with audited manual workflow.
History	A mismatch is finalized	Audit Log shows it above General Log; actor, ESI values, reason/version available.

21.1 Definition of done for the full-stack prototype

- Frontend uses live APIs/events instead of embedded clinical state, while preserving the current interaction design.
- All acceptance scenarios pass in CI and a replayable synthetic demo.
- Rules and model are independently versioned, deployable and rollback-capable.
- The audit trail reconstructs every comparison and override from immutable snapshots.

- Clinical, security, data and accessibility reviews have no unresolved severity-1 findings.
- A developer can run the prototype locally with documented seed data and no production credentials.

22. Delivery plan

Phase	Deliverables	Exit criteria
0 — Contract freeze	Schemas, rule config, event model, ESI labels, audit fields	Clinical/product/engineering approval
1 — Deterministic core	Data gateway, guardrail service, missing-data protocol, synthetic fixtures	All Layer 0 tests pass
2 — Queue and workflow	Assessment API, blind reveal, ranking, reassessment, history	End-to-end synthetic workflow passes
3 — Model baseline	Training pipeline, ordinal baseline, calibration, model registry, reason codes	Frozen evaluation and release gates pass
4 — Operations	Forecast service, integrations health, degraded modes	Forecast reproducible and comprehensible
5 — Shadow validation	FHIR sandbox/site mapping, observability, usability and clinical review	Go/no-go decision for controlled pilot

22.1 Ownership

Workstream	Accountable owner	Required reviewers
Product behavior	Product manager	Triage nurse, clinical informatics,

		frontend/backend
Clinical rules	Clinical safety lead	ED physician, nursing lead, informaticist
Model	ML lead	Clinical safety, data governance, fairness reviewer
Data/integration	Backend/data lead	EHR integration, security, clinical informatics
Human factors	Design/UX lead	Nurses, accessibility specialist, product
Audit/security	Security/compliance lead	Legal/privacy, backend, operations
Release	Engineering lead	All accountable owners

22.2 Key risks and mitigations

Risk	Impact	Mitigation
Undertriage	Delayed care/harm	Layer 0; asymmetric objective; adjudicated validation; human sign-off
Overtriage	Capacity strain	Calibrated thresholds; monitor overtriage; never relax critical safety to optimize flow
Automation bias	Nurse follows system uncritically	Blind nurse-first workflow; concise rationale; mismatch visibility

Alert fatigue	Warnings ignored	Acuity-linked schedules; deduplication; escalation policy; burden monitoring
Queue instability	Cognitive overload	Strict bands; bounded modifiers; motion rules; churn metric
Missing/bad data	Unsafe recommendation	Schema validation, provenance, abstention and manual verification
Site transportability	Performance varies by hospital	External/site-held-out tests, site configuration and shadow mode
Integration downtime	Stale context/capacity	Health/freshness states, conservative assumptions and downtime workflow
Regulatory misclassification	Deployment/legal risk	Early regulatory review tied to actual intended use and claims
Audit privacy	Excessive PHI exposure	Minimum necessary, scoped access, retention and access monitoring

22.3 Open decisions before coding beyond the prototype

- Who owns and approves the local ESI implementation and reassessment intervals?
- What dataset and adjudication process will define ground truth?
- What undertriage/overtriage thresholds constitute model release gates?
- What is the approved low-confidence abstention threshold and charge-nurse escalation policy?
- Which FHIR version/profile and hospital integration standards are available at the first target site?
- What constitutes a staffed treatment space at that site, and how frequently is it

refreshed?

- What audit retention and legal-hold policy applies?
- What exact regulatory claims will be made in the prototype/pilot?

Appendix A — Prototype configuration reference

Configuration	Current prototype default	Production requirement
Pediatric age boundary	<15 years	Clinical approval and versioning
Pediatric minimum SBP	$70 + 2 \times \text{age}$	Clinical approval; unit tests
Adult minimum SBP	90 mmHg	Clinical approval; unit tests
Critical SpO ₂	<90%	Clinical approval; device/source validation
Adult critical RR	<10 or >30/min	Clinical approval; boundary tests
Recheck intervals	5/15/45/90/180m for ESI 1–5	Site policy; configurable
Sepsis time modifier	(wait/60)×5 when temp >38.5 + tachycardia	Evidence/clinical review; bounded within band
Five-hour modifier	+50 at ≥300m if non-sepsis	Clinical/human-factors review
Decision window	75–135s adaptive	Usability validation; no auto action
Forecast horizon	60m at 15m steps	Versioned forecast assumptions
Simulation cadence	5 simulated minutes per 3.2	Test-only; isolated

	real seconds	environment
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Appendix B — Synthetic safety fixtures

Fixture	Key data	Expected result
Pediatric Control	3y; HR 130; SBP 85; SpO ₂ 98	Scored safely; no adult SBP/HR red flag
Arthur Hale	75y; HR 130; SBP 85; SpO ₂ 98	Critical age-adjusted SBP; ESI 1/top queue
John Doe	HR/BP/SpO ₂ null	Diagnostic Uncertainty; ATRIA abstains
Meera Shah	58y; temp 39.3; HR 132; wait 180m	Sepsis trajectory modifier and explanation
Aarav Kumar	72y; BP 82/58; SpO ₂ 87	Multiple Layer 0 reasons; ESI 1/top queue

Appendix C — Audit event minimum schema

JSON

```
{
  "event_id": "evt_01J...",
  "event_type": "NURSE_ATRIA_OVERRIDE",
  "occurred_at": "2026-08-27T14:12:10Z",
  "actor": {"id": "usr_...", "role": "TRIAGE_NURSE", "site": "site_..."},
  "subject": {"patient_id": "PT-1042", "encounter_id": "enc_..."},
  "links": {"snapshot_id": "snap_...", "assessment_id": "asm_..."},
  "decision": {"nurse_esi": 3, "atria_esi": 2, "final_action": "WAIT"},
}
```

```
"reason": {"code": "CLINICAL_JUDGMENT", "text": "..."},
"versions": {"ruleset": "...", "model": "...", "ui": "..."},
"integrity": {"sequence": 1842, "previous_hash": "..."}
}
```

Appendix D — Standards and reference sources

- [AHRQ — Emergency Severity Index overview](#)
- [HL7 FHIR R4 — Observation](#)
- [HL7 FHIR R4 — Encounter](#)
- [HL7 FHIR R4 — AllergyIntolerance](#)
- [HHS — HIPAA Security Rule](#)
- [HHS — HIPAA Audit Protocol and audit controls](#)
- [FDA — Clinical Decision Support Software guidance \(January 2026\)](#)

Internal product references:

- Digital_Ninjas.pdf — original Round 1 problem statement and four-layer concept.
- ATRIA_Intuitive_Flow_Deterioration_UI.html — current interaction and simulation reference build.

Appendix E — Glossary

Term	Definition
ATRIA	Clinical decision-support workstation that recommends ESI and continuously orders attention.
Attention rank	Dynamic queue position; not the same as clinical ESI.
Blind assessment	Nurse selects ESI before ATRIA's recommendation is revealed.
Critical undertriage	A critical/emergent patient is assigned a less urgent level.
Diagnostic Uncertainty	Essential inputs are missing or invalid; ATRIA

	abstains.
Guardrail	Deterministic safety rule that operates before and independently of ML.
Harvey ball	Non-numeric visual indication of the decision window.
Override	Final nurse decision is less urgent than ATRIA, conflicts with a red flag, or proceeds after abstention.
Reassessment	A new clinical review/vital set for a patient who remains waiting.
Safety band	Ordering group that prevents operational modifiers from crossing ESI urgency boundaries.
Staffed space	A treatment space that is physically available and safely staffed, not merely a licensed bed.

Final implementation principle

When product optimization, operational pressure and clinical safety conflict, preserve the red-flag gate, missing-data abstention, nurse authority and complete audit trail. Optimize the workflow around those invariants—not through them.